#ManyDaughters—Pre-analysis Plan

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Abstract

In this document, we describe our pre-analysis plan (PAP) for the *#ManyDaughters* research project. The PAP will contain a brief introduction to motivate and justify the choices we make for our empirical analysis. Then we describe for each of the 4 hypotheses how the dependent and explanatory variables are defined, which sample period we consider, and how the coefficient of interest should be interpreted. Besides the regression equation, we will also include the exact Stata command to estimate the regression. Our research team (RT) was randomised into the group with no data access while writing the PAP.

1 Introduction

The #ManyDaughters project seeks to analyse whether having daughters affects preferences and realised household outcomes. Before describing in the next section how exactly each of the four hypotheses is tested, this section will describe how we define the treatment and control group, the main identifying assumption behind our analyses, and which samples we will consider.

Our interpretation of the hypotheses of interest is to compare outcomes between parents of only daughters to those having only sons (multiple children or one child). An alternative control group would be individuals who not (yet) have children at all. But we believe that this is a less valid control group, as having a child affects many outcomes regardless of the child's gender (e.g. labour market outcomes, division of unpaid work, sleep, attitudes, see e.g. Cortés and Pan, 2023; Jessen, 2022; Kleven et al., 2024; Kuziemko et al., 2018; Richter et al., 2019) and because parents differ from childless individuals in many observable and unobservable characteristics.

Our key identifying assumption is that the gender of a child is exogenous with respect to potential outcomes ($E[u_i|X] = 0$).[1] An often cited "natural ratio" is that for 100 girls around 105 boys are born (The Economist, 2025) as boys have a slightly lower survival rate than girls. This also appears to be

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[1] Recent advances in IVF technology allow to select the sex of babies, but this practice is forbidden in many countries and the technology was not available in Germany in the time period we are considering. Furthermore, the number of IVF births in Germany in our sample period is relatively low with a share of around 2%.

the case in Germany, as in 2023 5% more boys than girls were born (Destatis, 2025). Selective abortions due to the sex of the fetus are also forbidden in Germany according to §15 of the *Gendiagnostikgesetzes (GenDG)*, which states that for this reason the sex of the fetus can only be reported to parents after the 12th week of pregnancy. If the sex of children is exogenous (with respect to potential outcomes), the comparison of parents with daughters or sons should resemble a controlled experiment in which either group is identical in characteristics. The useful feature of this is that panel data are not required to test the hypotheses and a simple cross-sectional regression suffices.

To cleanly assign parents to be in the control or treatment group, we focus on parents whose children have the same sex. This could either be those with only one child or with multiple children of the same sex. In 2022, among mothers aged 45-49 where fertility is mostly completed, 30% had one child, 47% had two children, and 23% had more than three children (Destatis, 2024). All one-child families will necessarily enter our sample and this should also be the case for slightly more than 50% of two-child families and around 25% of three-child families etc. given the assumed randomness of child sex.

To allow for a comparison between the four hypotheses we are testing, we are imposing some joint sample restrictions for all analyses which are guided by data availability or the required sample structure for some analyses. In our analyses, we consider:

I. Two-parent different sex households. *H3* requires to look at within-household outcomes, and as we test for gender differences, we require parents to be of a different sex.

II. We only consider biological children for treatment assignment.

III. Households with youngest child aged 0 to 5 years. This is the age group for which the dependent variables of *H1* and *H4* are available in SOEP and it matches the *Familien in Deutschland (FiD)* L1 sample (Geue et al., 2020) these question are addressed at.

IV. Use the years 2005 to 2019 in the analysis, if the data is available. The dependent variable in *H2* is available in four survey waves in this time period. Other dependent variables are available in every year, some only in one year. This will be described for each hypothesis in more detail.

V. All SOEP samples for which all variables (dependent and independent) are non-missing.

VI. Include individual-level survey weights **phrf** in all analyses.

VII. Standard errors are clustered at the household-level for *H1*, *H2* and *H4*, and we use robust standard errors for the household-level estimation in *H3*.

2 Testing the Hypotheses

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

The best measures of attitudes towards gender inequality in SOEP (Goebel et al., 2019; Siegers et al., 2021) were asked in the individual *Familien in Deutschland (FiD)* questionnaire in 2012. Specifically, we

will construct an equally weighted index consisting of the following three variables:

1. **plh0303:** Men should contribute to housework just as much as women.
2. **plh0307:** Men should take greater responsibility for the financial support of the family than women.
3. **plh0308_v2:** The best situation is when both the man and the woman are equally employed and both share household and family responsibilities equally.

These variables are contained in the data set *pl*. Each variable contains around 7000 non-missing observations in 2012. The values of the variables range from 1 (fully agree) to 4 (fully disagree). The index will be created for all observations containing non-missing values for all three items. Specifically, we will first sum up the indices such that higher values denotes more support for gender equality:

$$index_H1 = (5 - \text{plh0303}) + \text{plh0307} + (5 - \text{plh0308_v2})$$

$index_H1$ ranges from 3 to 12 with higher numbers implying more support for gender equality. This index will be standardised to have a mean of 0 and a standard deviation of 1 in our sample. The standardized index will be called std_index_H1 and constitutes the dependent variable to test *H1*.

We chose these three measures of gender attitudes to construct the index as they explicitly elicit support for gender equality. Other related outcomes are often used in the literature as proxies for gender inequality (e.g., **plh0302_v1** asking whether a child below age 3 suffers if the mother is working), but are not directly asking for support of gender equality. An issue with several related variables is that they are not asked in a gender neutral way (e.g., for the variable **plh0302_v1** no identically phrased question about fathers is asked) such that they are not suitable as measures of gender equality. We identified no other questions related to gender attitudes that explicitly ask for support for gender equality.

As described in the first section, the sample will consists of all household fulfilling the listed criteria in that section and with non-missing values for the following control variables (SOEP data set containing the data is reported in parentheses):

— **age_first_birth:** Age at birth of first child calculated using the variables **gebjahr** (own birth year) and **syear** (survey year) in the *ppathl* data and information on the birth year and month of the first child (**kidgeb01** and **kidmon01** contained in *biobirth*)

— **sex (ppathl)**: sex of the parent

— **migback (ppathl)**: migration background

— **l11101 (pequiv)**: federal state

— **pgpsbil (pgen)**: highest schooling degree

— **num_child**: number of children in the survey year, calculated using the information on all children from the same *biobirth* data as in the first bullet point

— **age_old_child**: Age of the oldest child similarly calculated with *biobirth* data; this variable and **age_first_birth** will be measured in full years

The control variables are not required from an identification perspective, but could increase the precision of estimates.

We estimate the following regression:

$$std_index_H1_{ij} = \alpha_1 + \alpha_2 \text{DAUGHTER}_j + \alpha_3 X_i + \alpha_4 \lambda_j + \epsilon_{ij} \quad (1)$$

X_i is a vector of control variables listed above of an individual i and λ_j contains the controls at the level of the household j, whereby each variable is coded as a dummy variable for each category. Standard errors are clustered at the household-level. The indicator variable DAUGHTER is equal to 1 if all children in the household are girls and 0 if all children are boys. The coefficient of interest is α_2 and denotes how the standardized index of gender attitudes is affected by having only daughters (vs. having only boys).

We will estimate the regression with Stata using the following command:

reg std_index_H1 daughters i.age_first_birth i.sex i.migback i.l11101
i.pgpsbil i.num_child i.age_old_child [aw=phrf], vce(cluster hid)

The coefficient of the variable *daughters* is α_2. The point estimate, standard error, p-value and number of observations in the estimation (in that order) will be extracted as follows:

— global coef_H1: dis %5.4fc= r(table)[1,1]

— global se_H1: dis %5.4fc= r(table)[2,1]

— global pval_H1: dis %5.4fc= r(table)[4,1]

— global N_H1: dis %13.0fc= e(N)

H2. Having daughters leads to more left-leaning political preferences.

To test left-leaning political preferences, we will again use a direct measure of the outcome of interest rather than an imperfect proxy (such as voting behaviour, preferences for redistribution, support for certain policies etc.). Our dependent variable will be a recoded version of:

— **plh0004** which asks respondents on a scale from 0 (very left leaning) to 10 (very right leaning) about their political preferences. The variable is contained in the data set *pl*.

The variable **plh0004** is included in the SOEP in 2005, 2009, 2014 and 2019, and we will use all 4 years in the estimation. To ensure that the hypotheses is tested directly, we recode the variable such that higher values means more left leaning: $left_pref = 11 - \text{plh0004}$ and we again standardise the variable to have a mean of 0 and a standard deviation of 1. The standardised variable will be called $std_left_pref_H2$.

The control variables used in the estimation will be the same as in *H1* but we additionally include survey year FEs. The otherwise similar regression equation is:

$$std_left_pref_H2_{ijt} = \beta_1 + \beta_2 \text{DAUGHTER}_{jt} + \beta_3 X_{it} + \beta_4 \lambda_{jt} + year_t + \epsilon_{ijt} \quad (2)$$

where the subscript t indicates the different survey years. The coefficient of interest is now β_2 and we additionally include survey wave fixed effects in the analysis. We will estimate the regression with Stata using the following command:

reg std_left_pref_H2 daughters i.age_first_birth i.sex i.migback i.l11101
i.pgpsbil i.num_child i.age_old_child i.syear [aw=phrf], vce(cluster hid)

Relevant statistics will be extracted after estimation similarly to the procedure in *H1*.

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Gender equality in the household measured through behaviours or outcomes could mean a range of things. The most natural interpretation is that it refers to equality in time spent conducting certain activities, such as housework, child care or paid work. We believe that focusing on one of these items would be selective and it is more desirable to have a single measure combining them.

As the dependent variable for this hypotheses, we will use the first specialisation index proposed by Siminski and Yetsenga (2022)[2]:

$$SI = 1 - |\frac{DW_F}{DW_F + DW_M} - \frac{MW_F}{MW_F + MW_M}|$$

DW refers to (unpaid) domestic work, which combines errands, housework, childcare, care for other people, and repairs and gardening. MW is time spent on (paid) market work. The subscripts refers to men or women. The index ranges from 0 to 1, where 0 means that households fully specialise and 1 means no specialisation at all (i.e., market and domestic work are similarly distributed). Note that this index is gender-blind, but in practise the vast majority of households specialise towards women being responsible for more domestic work even when market work is equally shared (Jessen et al., 2024).

The SOEP questionnaire asks for time spent on a number of activities on a regular weekday (every survey year) and on weekend days (every other year). We use answers to the question *"What is a typical day like for you? How many hours do you spend on the following activities on a typical weekday?"*

We focus on weekdays and calculate domestic work (DM) by summing up the following items:

— **pli0040**: errands on weekdays

— **pli0043_h**: housework on weekdays

— **pli0044_h**: childcare on weekdays

— **pli0046**: care for people in need on weekdays

— **pli0049_h**: repairs and gardening on weekdays

Market work (MW) is measured through the variable:

[2]We directly recode their first index such that, in line with the hypothesis, a value of 1 indicates no specialisation.

— **pli0038_v4**: time spent working (including travel and secondary jobs)

All these variables are contained in the *pl* data set. The different activities can potentially add up to more than 24 hours, and especially for childcare some parents report 24 hours or more spent on this activity (see Appendix in Jessen, 2022). We restrict domestic work and market work to not exceed 20 hours per weekday together. In this process, we preserve the original shares. For example, if a respondent indicates 15 hours of domestic work and 15 hours of market work, we would restrict both to 10 in our censored version. We hence still keep all individuals that have responded to the question (i.e. we winsorise the values but drop no observations). We use all years in our sample period for this analysis, i.e. 2005 to 2019.

We again standardise the index *SI* to have a mean value of 0 and a standard deviation of 1, and we call this index *std_SI_H3*. The estimation is conceptually similar to the previous estimations, but it is now conducted at the household level. Individual-level control variables (education, age at first birth) can still be included in the regression, but standard errors are now robust and not clustered at the household level.

$$std_SI_H3_{jt} = \gamma_1 + \gamma_2 \text{DAUGHTER}_{jt} + \gamma_3 X_{it} + \gamma_4 \lambda_{jt} + year_t + \epsilon_{ijt} \tag{3}$$

The coefficient of interest is γ_2. X_{it} contains individual-level controls and λ_{jt} control variables at the household-level (age of oldest child, number of children, federal state).

The Stata regression reads as follows, the subscript m indicates controls for the mother, f stands for fathers:

reg std_SI_H3 daughters i.age_first_birth$_m$ i.age_first_birth$_f$ i.migback$_m$ i.migback$_f$ i.l11101 i.pgpsbil$_m$ i.pgpsbil$_f$ i.num_child i.age_old_child i.syear [aw=phrf], robust

Relevant statistics are extracted as above.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

H4 is a test of a heterogeneous effect of *H1* by parents' sex. We use essentially the same specification as in *H1*, but interact the treatment dummy (DAUGHTER) with the sex of the parent.

$$std_index_H1_{ij} = \delta_1 + \delta_2 \text{DAUGHTER}_j + \delta_3 \text{DAUGHTER}_j \times \text{MAN} + \delta_4 X_i + \delta_5 \lambda_j + \epsilon_{ij} \tag{4}$$

The indicator variable MAN equals 1 if the respondent is a man (corresponds to value 2 of the variable **sex**). The coefficient of interest is δ_3 and denotes whether the effect on gender attitudes differs between men and women.

We will estimate the regression with Stata using the following command:

reg std_index_H1 daughters##man i.migback i.age_first_birth i.l11101

i.pgpsbil i.num_child i.age_old_child [aw=phrf], vce(cluster hid)

The statistics of interest are stored in the 8th column and extracted as described in H1:

— global coef_H4: dis %5.4fc= r(table)[1,8]

— global se_H4: dis %5.4fc= r(table)[2,8]

— global pval_H4: dis %5.4fc= r(table)[4,8]

— global N_H4: dis %13.0fc= e(N)

3 Interpretation of Coefficients

The analyses are conducted in a similar way, allowing for direct comparison of reported coefficients. All four hypotheses have a standardised dependent variable (mean 0, standard deviation 1). The main explanatory variable is a dummy variable (or for *H4* an interaction of two dummy variables).

Because we code the dependent variables the way the hypotheses are phrased, the point estimates with the standard error (or p-value) will directly indicate whether the hypotheses are true according to our analysis. In each case, a positive and statistically significant coefficient means that we find support for the respective hypothesis. Out of convention, we declare support for a hypothesis if a positive coefficient is statistically significant at the 5% level ($p - value < 0.05$). In contrast, a negative or statistically insignificant coefficient means that we reject the respective hypothesis.

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